MIL-F-81966(AS)
19 April 1974
Superseding
AS-1688C
28 September 1971

MILITARY SPECIFICATION

FUZE FUNCTION CONTROL SET AN/AWW-4(V)

This specification has been approved by the
Naval Air Systems Command, Department of the Navy.

1. SCOPE

1.1 Scope - The equipment covered by this specification shall be for airborne use and shall provide four pilot selectable d-c voltage options of two levels and two polarities for use with electric bomb and missile fuzes.

1.2 Classification - The equipment covered by this specification shall consist of the following items:

Type Designation	Applicable Paragraph
PP-6419/AWW-4(V)	3.5.1
C-8441/AWW-4[V)	3.5.2
C-8442/AWW-4(V)	3.5.2
MX-8556/AWW-4(V]	3.5.3
MX-8557/AWW-4(V)	3.5.4
MX-8558/AWW-4(V)	3.5.5
MX-9169(AWW-4[V)	3.S.6

1.3 Associated Equipment - This equipment shall operate with the associated equipment listed in 6.6.

2. APPLICABLE DOCUMENTS

2.1 General - The following documents of the issue in effect on the date of invitation for bids form a part of this specification to the extent specified herein:

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SPECIFICATIONS

Military

MIL-E-5400	Electronic Equipment, Aircraft, General Specification for
MIL-T-5422	Testing, Environmental, Aircraft Electronic Equipment
MIL-T-5442	Thread Compound, Antiseize and Sealing, Oxygen Systems
MIL-I-6181D	Interference Control Requirements Aircraft Equipment
MIL-E-17555	Electronic and Electrical Equipment, Accessories and Repair Parts; Packaging and Packing of
MIL-T-18303	Test Procedures; Preproduction, Acceptance and Life, for Aircraft Electronic Equipment, Format for

STANDARDS

Military

MIL-STD-105	Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-704	Electric Power, Aircraft Characteristics and Utilization of
MIL-STD-781	Reliability Tests, Exponential Distribution
MIL-STD-794	Parts and Equipment, Procedures for Packaging and Packing of
MIL-STD-810	Environmental Test Methods

DRAWINGS

Naval Air Systems Command

DL 67A24F1	Data List for Fuze Function Control Set AN/AWW-4(V)
DL 125AS1000	Data List for PP-6419/AWW-4(V)
DL 126AS100	Data List for C-8441/AWW-4(V) and C-8442/AWW-4(V)
DL 67A21D1	Data List for MX-8557/AWW-4(V) and MX-8558/AWW-4(V)
DL 67A22D1	Data List for MX-8556/AWW-4(V)
DL 203AS100	Data List for MX-9169/AWW-4(V)
DL 67A35F1	Data List for MT-3804\AWW-2A

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PUBLICATIONS

Military

ANA Bulletin No. 400	Electronic Equipment; Aircraft and Guided Missiles, Applicable Documents

2.1.1 Availability of Documents -

(1) When requesting specifications, standards, drawings, and publications refer to both title and number. Copies of this specification and applicable specifications and documents required by contractors in connection with specific procurement functions may be obtained upon application to the Commanding Officer, Naval Supply Depot, Code 105, 5801 Tabor Avenue, Philadelphia, Pennsylvania 19120.

3. REQUIREMENTS

3.1 Preproduction - This specification makes provision for preproduction testing.

3.2 Parts and Materials -

3.2.1 General Requirements - The parts and materials specified in the drawings and related documents for use in construction of this equipment were selected to ensure the fulfillment of all requirements of this specification. Any deviation by the contractor from the use of these specified parts and materials necessarily places both the burden of proof for the acceptability of the change, and the cost associated therewith, upon the contractor. Approval for deviations must be requested from the procuring activity by the contractor and granted by the procuring activity prior to submission of equipments for preproduction tests. Approval of parts and material deviations shall not be construed by the contractor to constitute a waiver of the performance requirements or any other requirements of this specification.

3.2.1.1 Nonstandard Parts and Material Approval - Approval for the use of nonstandard parts and materials shall be obtained by the contractor as outlined in MIL-E-5400.

3.2.2 Supplemental Requirements - In the event that the contractor proposes to alter the equipment design delineated by this specification and applicable drawings, the following supplemental requirements for the selection of parts and materials are applicable.

3.2.2.1 Basic Conformance - Parts and materials selected for use in the equipment shall conform to the requirements of MIL-E-5400. parts shall be selected from those parts included in ANA Bulletin No. 400, unless the parts listed therein will not perform adequately in the intended applications and environments specified herein.

3.2.2.2 Interchangeability - The equipment shall meet the interchangeability requirements of MIL-E-5400. The drawings listed in DL 67A24F1 and related lists shall be the standard for measuring interchangeability of all applicable units, assemblies, subassemblies, and parts.

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3.3 Design and Construction - The equipment shall be manufactured in accordance with the drawings and subsidiary specifications listed in DL 67A24F1 and related lists, except insofar as errors of omission or commission on the drawings may be incompatible with the requirements of this and related specifications. In any case it is a condition of final acceptance that the equipment manufactured meet all the performance requirements of this specification, and such acceptance test documents as may be applicable. Proposed changes, either to correct drawing discrepancies or to effect improvements, shall be specifically approved by the procuring activity prior to their incorporation.

3.3.1 Total Weight - The total weight of the equipment shall be as specified in the detail equipment requirements. See paragraph 3.5.

3.3.2 Reliability -

3.3.2.1 Operating Life - The equipmuent shall have a minimum total operating life of 10,000 hours with reasonable servicing and replacement of parts.

3.3.2.2 Reliability in Mean-Time-Between-Failure (MTBF) - The equipment shall have a specified bean (operating) time between failures of 1000 hours as defined in MIL-STD-781 when tested and accepted in accordance with 4.4.3 of this specification.

3.3.3 Interference Control - The generation of radio interference by the equipment and the vulnerability of the equipment to radio interference shall be within the limits of MIL-I-6181D during steady state no load operation. This requirement does not apply during switching operations or when the system is pickled.

3.3.4 Standard Conditions - The following conditions shall be used to establish normal performance characteristics under standard conditions and for making laboratory bench tests.

Temperature	Room ambient (25 ±5°C)
Altitude	Normal ground
Vibration	None
Humidity	Room ambient up to 90% relative
Input power voltage	115 ± 1.0 volts, 400 ± 5 CPS a-c and 27.5 ± 0.5 volts d-c

3.3.5 Service Conditions - The equipment shall operate satisfactorily under any of the environmental service conditions or reasonable combination of these conditions as specified in MIL-E-5400 for class 2 equipment, except as modified herein.

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3.3.5.1 Tests - The Fuze Function Control Set AN/AWW-4(V) shall pass the following tests of MIL-E-5400 as detailed in MIL-T-5422:

a. Temperature-Altitude for Class 2 equipment steps 1 through 6 only except that the 4 hours, 71°C temperature exposure at ambient pressure of step 6 shall be modified to 4 hours, 71°C, at 70,000 feet altitude

b. Vibration

c. Shock

d. Humidity

e. Salt Fog

f. Explosion per test procedure I

g. Fungus

The tests specified shall be conducted in the order specified in MIL-T-5422. The equipment shall be operated during testing where specified in MIL-T-5422 and meet the applicable performance requirements of 3.4. Equipment performance shall be verified at the completion of each test.

3.3.6 Warm-Up Time - The time required for the equipment to warm up prior to operations shall not exceed 1 minute under standard conditions and 2 minutes at extreme service conditions. No warm-up time shall be permitted prior to performing the voltage drift test of 3.4.9.

3.3.7 Input Electrical Power -

3.3.7.1 Operating Power - The equipment shall meet all applicable requirements of MIL-STD-704 and shall give specified performance when energized from the following power sources having characteristics and limits as defined in MIL-STD-704 and herein. The power required shall not exceed the specified amounts.

(1) AC Power (Three Phase), 115/200V, Category B, 1000 VA

(a) Extreme limits: 106 to 120 volts a-c line to neutral, 370 to 430 Hz

(2) DC Power, 28V, Category B, 15 Watts

(a) Extreme limits: 24 to 30 volts d-c

3.3.7.2 Lighting Power - Input power for lighting shall require not more than 1 amp at 5 volts or 0.5 amps at 28 volts.

3.3.7.3 Degraded Performance - Degraded performance will be permitted for voltage transients not exceeding 0.5 second during normal electrical system operation. Operation shall return to normal with-no resulting damage to the equipment.

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3.4 Performance - The performance of the equipment shall be as set forth below plus any and all performance requirements stated on the drawings of DL 67A24F1 and related lists. Unless otherwise specified, these requirements shall apply under both standard and extreme service conditions. When reduced performance requirements for service conditions are acceptable, such variations are specified. Unless otherwise noted, the performance is specified with the Control Box and Power Supply interconnected, operating power applied and the wheels interlock circuit closed in accordance with the test circuit of figure 1. The high voltage output of the power supply appears across pins A and B of 1J2 only when the pickle switch is closed (see 3.4.12).

3.4.1 No Load Voltage Adjustments at Standard 0perating Power Input. These adjustments shall be made only once and only at the beginning of the series of preproduction or acceptance tests listed in table 1. The equipment shall be permitted to warm-up and stabilize prior to making these adjustments. All applicable test requirements shall be met without further adjustment. The equipment shall not be readjusted for each environmental exposure after the initial adjustments have been made.

3.4.1.1 VT Position - With the Fuze Function Control switch in the VT position the power supply output shall be adjusted to +300 ± 1.0 volts d-c under standard conditions.

3.4.1.2 INST Position - With the Fuze Function Control switch in the INST position the power supply output shall be adjusted to +195 ± 1.0 volts d-c under standard conditions.

3.4.2 No Load Voltage at Standard Operating Power Input -

3.4.2.1 VT Position - With the Fuze Function Control switch in the VT position, the power supply output shall be within +1.0 volt of 3.4.1.1 at 25 + 10°C. At all other temperatures the output shall be within + 2.0 volts of 3.4.1.1.

3.4.2.2 DLY 2 Position - When the Fuze, Function control switch is switched from the VT position to the DLY 2 position, the polarity of the power supply output must be negative and the magnitude of the output voltage shall not vary more than +0.5 volts from the voltage of 3.4.2.1.

3.4.2.3 INST Position - With the Fuze Function Control set in the INST position, the power supply output shall be within ± 0.8 volt of 3.4.1.2 at 25 ± 10°C. At all other temperatures the output shall be within ± l.O volt of 3.4.1.1.

3.4.2.4 DLY 1 Position - When the Fuze Function Control switch is switched from the INST position to the DLY 1 position, the polarity of the power supply output must be negative and the magnitude of the output voltage shall not vary more than ± 0.5 volt from the voltage of 3.4.2.3.

3.4.3 No Load Output Voltage at Extreme Limits of A-C 0perating Power Inputs -

3.4.3.1 With an operating power input of low line voltage and low line frequency, the power supply output shall be as follows:

a. VT Position - The output shall be within ± l.O volt of 3.4.2.1

b. DLY 2 Position - The output shall be within ± l.O volt of 3.4.2.2

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c. INST Position - The output shall be within +0.5 volt of 3.4.2.3

d. DLY 1 Position - The output shall be within +0.5 volt of 3.4.2.4

3.4.3.2 with an operating power input of high line voltage and low line frequency, the power supply output shall be as specified in 3.4.3.1 a, b, c, and d.

3.4.3.3 With an operating power input of low line voltage and high line frequency, the power supply output shall be as specified in 3.4.3.1 a, b, c, and d.

3.4.3.4 With an operating power input of high line voltage and high line frequency, the power supply output shall be as specified in 3.4.3.1 a, b, c, and d.

3.4.4 Recovery Time at Standard A-C Operating Power under Varying Loads -

3.4.4.1 VT Position [MK-344} - With the Fuze function Control switch in the VT position, the output voltage shall recover to within 3.0 volts d-c of the no-load value within 6 milliseconds after application of a simulated 6 fuze {MK-344) load shown in figure 2B.

3.4.4.2 INST Position MK-344) - With the Fuze Function Control switch in the INST position, the output voltage shall recover to within 2.0 volts d-c of the no-load value-within 4 milliseconds after application of a simulated 6 fuze (Mk-344) load shown in figure 2B.

3.4.4.3 VT Position (Mk-344) - Repeat 3.4.3.1 except that the load (1 fuze Mk-344) shall be as shown in figure 2A and the output voltage shall recover to within 3.0 volts d-c of the no-load value within 1 millisecond.

3.4.5 Recovery Times Under Varying Loads at Extreme Limits of A-C Operating Power Inputs -

3.4.5.1 Repeat 3.4.4.1 through 3.4.4.3 except that the a-c operating power input shall be as follows:

a. Low line voltage and low line frequency

b. High line voltage and low line frequency

c. Low line voltage and high line frequency

d. High line voltage and high line frequency

3.4.6 Output Voltage Under Constant Load at Standard Operating Power Input -

3.4.6.1 VT Position - With the Fuze Function Control switch in the VT position, the output shall be within ± 1.5 volts of 3.4.2.1 at 25 ± lO°C when a 1000 ± 10 ohms load is applied. At all other temperatures the output shall be within ± 3.O volts of 3.4.2.1.

3.4.6.2 INST Position - With the Fuze Function Control switch in the INST position, the output shall be within ± 1.0 volts of 3.4.2.3 at 25 ± 10°C when a 1000 ± 10 ohms is applied. At all other temperatures the output shall be within ± 2.O volts of 3.4.2.1.

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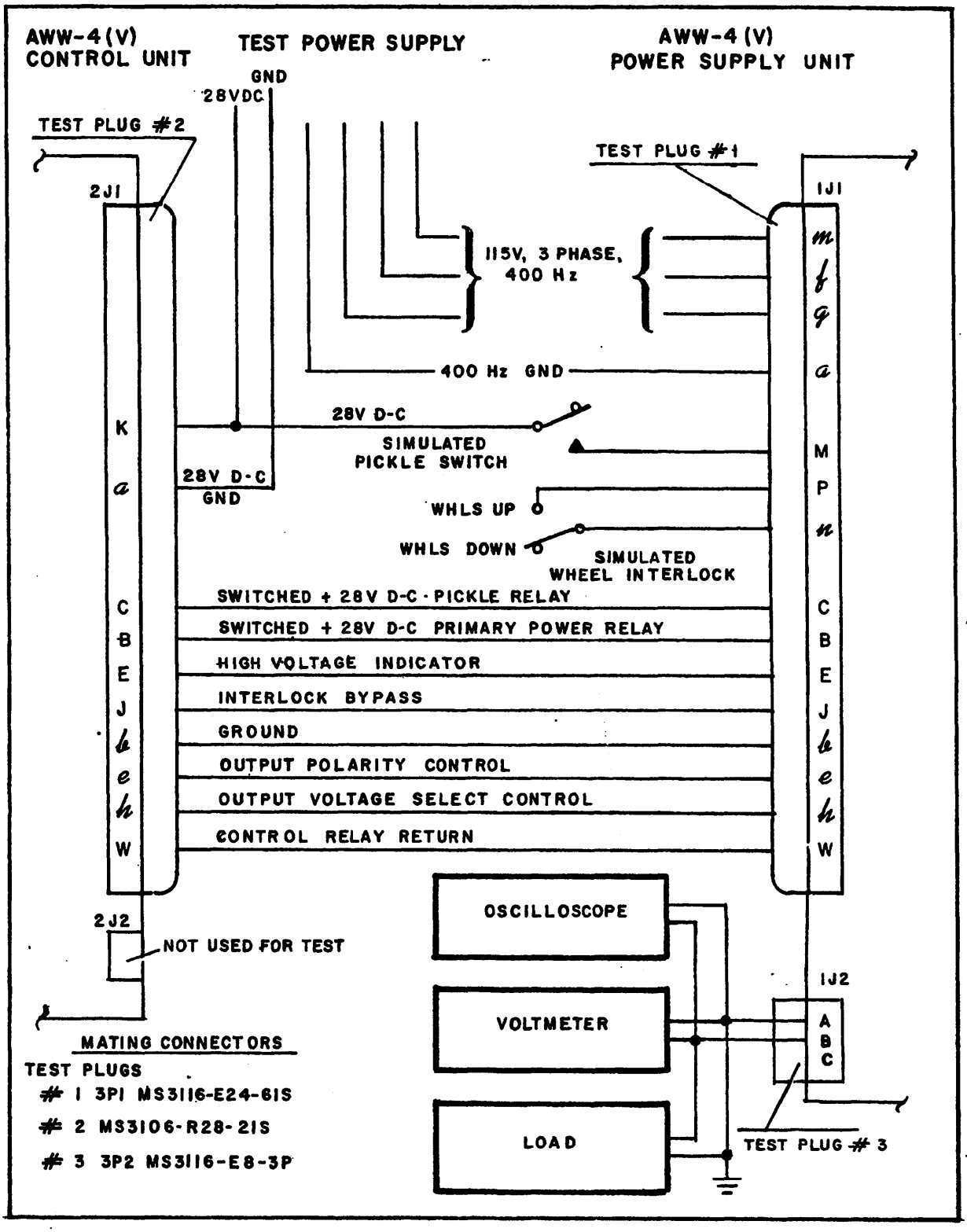


Figure 1. AWW-4 Typical Test Circuit

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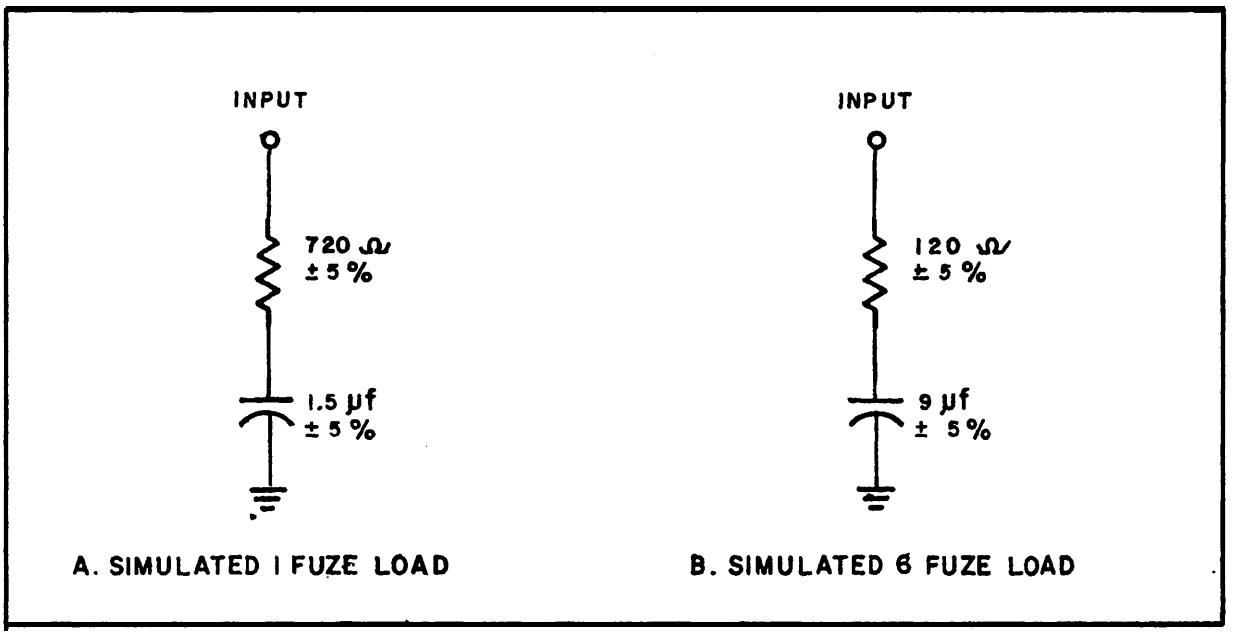


Figure 2. Simulated Mk-344 Loads

3.4.7 Output Voltage Under Constant Load at Extreme Limits of A-C Operating Power Inputs -

3.4.7.1 Repeat 3.4.6.1 and 3.4.6.2 with the operating power inputs specified in 3.4.5.1 a, b, c, and d. The magnitude of the output voltage shall not vary more than ± 0.5 volts from the voltages of 3.4.5.1 and 3.4.5.2.

3.4.8 Short Circuit at Standard Operating Power Input -

3.4.8.1 With the Fuze Function Control switch in the VT position, the power supply output shall be subjected to a 2 ohm + 10 percent short circuit for a duration of 5 + 1 seconds. After removal of-the short, the output voltage shall be within +0.5 volt of the output voltage prior to the application of the short circuit.

3.4.8.2 With the Fuze Function Control switch in the INST position, repeat 3.4.8.1. After removal of the short, the output shall be within ± 0.5 volt of the output voltage prior to the application of the short circuit.

3.4.9 Short Circuit at Extreme Limits of A-C Operating Power Inputs -

3.4.9.1 Repeat 3.4.8.1 and 3.4.8.2 with the operating power inputs specified in 3.4.5.1 a, b, c, and d.

3.4.10 Voltage Drift at Standard Operating Power Input -

3.4.10.1 VT Position - With power supply at room temperature and standard operating power input applied, the Fuze Function Control switch shall be switched to the VT position. The open circuit output voltage shall not drift more than +1.50 volts d-c within 30 minutes after turn-on. Voltage readings shall be taken

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immediately after turn-on, then each minute for the first 5 minutes and every 5 minutes thereafter. No warm-up time shall be permitted prior to this test.

3.4.10.2 INST Position - Repeat 3.4.10.1 except the Fuze Function Control switch shall be in the INST position and the maximum voltage drift shall be ± 1.0 volts d-c.

3.4.11" INTERLOCK Operation - When a Fuze Function Control switch position, other than SAFE is selected, the primary power input relay of the power supply shall be energized to apply a-c power to the power supply circuitry only if an external "wheels interlock" circuit path is completed between pins 1J1-n and 1J1-p or the interlock bypass circuit is activated by depressing the HVDC TEST switch on the control box.

3.4.12 PICKLE Circuit Operation-

3.4.12.1 Activation Time - The output voltage (in accordance with the function switch position selected all appear at 1J2-B within 20 milliseconds of the "application of 28 Vd-c [pickle voltage) to pin lJl-M of the power supply and shall remain as long as-the 28 V is applied to pin M.

3.4.12.2 Delay Time - The output voltage shall remain at 1Jz-B for 0.7 to 2.0 seconds after the voltage applied to pin 1J1-M is removed.

3.4.13 High Voltage Indication -

3.4.13.1 Turn On Level - The power supply shall establish a ground path through pin 1J1-B to complete the circuit which turns on the HVDC indicator lamp on the control box when the high voltage d-c generated in the power supply exceeds 93 percent of:

a. +300 volts with FFC switch in the VT position

b. +195 volts with FFC switch in the INST position

c. -195 volts with FFC switch in the DLY1 position

d. -300 volts with FFC switch in the DLY2 position

3.4.13.2 Turn Off Level - The HVDC lamp on the control box shall not illuminate when the high voltage generated in the power supply is less than 75 percent of the levels specified in 3.4.13.1.

3.4.14 Resistance at Output - With the pickle circuit de-energized {no HVDC out of 1J2) the d-c resistance between pins 1J2-A and 1J2-B shall be 470 ohms + 15 percent.

3.5 Detail Requirements -

3.5.1 Power Supply PP-6419/AWW/-4[V) -The Power Supply PP-6419/AWW-4(V) shall be manufactured in accordance with the documents listed on DL 125AS1OOO and shall meet the requirements stated thereon and in the following subparagraphs.

3.5.1.1 Function - The Power Supply PP-6419/AWW-4(V) converts 3 Ø 400 Hz power *to* ± 300 and ± 195 volts d-c for use with electric bomb and missile fuzes. Any

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one of these four regulated d-c voltages can be selected by using the control signal from Control Box C-8441 or C-8442/AWW-4(V) or their equivalent. Other external signals activate an a-c power interlock relay and a relay that applies the selected voltage to the output connector. The power supply also provides a signal to illuminate a light on the cockpit control to indicate the presence of the high voltage in the power supply. The power supply, when actuated using the C-8441 or C-8442 control box or equivalent, shall meet the requirements of 3.4.

3.5.1.2 Weight - the weight of the Power Supply PP-6419/AWW-4(V) shall not exceed 6.5 pounds.

3.5.1.3 Content - The power supply contains the following assemblies and associated circuits to develop, regulate, select and apply the proper voltage to the output.

(1) Regulator Printed Wiring Board Assembly 1A1

(2) Sense/Pickle Printed Wiring Board-Assembly 1A2

3.5.1.3.1 Regulator Printed Wiring Board Assembly 1A1 - The Regulator Printed Wiring Board Assembly lA1 responds to a sample of the output voltage and drives the output power transistors to maintain a regulated output voltage.

3.5.1.3.2 Sense/Pickle Printed Wiring Board 1A2 - The Sense/Pickle Printed Wiring Board 1A2 senses the magnitude of the output voltage and provides a signal to illuminate a light in the cockpit control to indicate when the voltage is within prescribed limits. This assembly also contains the circuitry that drives the output relay when a "pickle" or release pulse is applied to the power supply and provides for a short holding delay after the release pulse is received.

3.5.2 Controls C-8441 and C-8442/AWW-4(V) - The Controls C-8441 and C-8442/AWW-4(V) shall be manufactured in accordance with the documents listed on DL 126AS100 and shall meet the requirements stated thereon and in the following subparagraphs.

3.5.2.1 Function - The Controls C-8441 and C-8442/AWW-4[V) are installed in an aircraft cockpit and are used by the pilot to provide the signals that select the voltage of the Power Supply PP-6419/AWW-4(V). The controls contain a HVDC lamp that is turned on when the power supply is producing high voltage and a HVDC TEST button to facilitate ground checkout of the AFJ/AWW-4(V) system when installed in the aircraft. The two controls are identical except for the illuminating panel on the face of the controls. The C-8441/AWW-4(V) contains an illuminating panel with imbedded bulbs that utilize 28 volt power for installation in aircraft with 28 volt lighting power. The C-8442/AWW-4(V) illuminated panel contains 5 volt bulbs for use in aircraft with 5 volt lighting power. The C-8441 and C-8442/AWW-4(V) controls shall operate Power Supply PP-6419/AWW-4(V) and meet the requirements of 3.4.

3.5.2.2 Weight - The weight, of the Control C-8441 C-8442/AWW-2(V) shall not exceed 2.5. pounds.

3.5.3 Adapter, Mount MX-85S6/AWW-4(V) - The Adapter Mount MX-8556/AWW-4(V) shall be manufactured in accordance with the documents listed on DL 67A22D1 and shall meet the requirements stated thereon and in the following subparagraphs.

3.5.3.1 Function- The Adapter Mount MX-85S6/AWW-4(V) provides the mechanical interface used to adapt the Power Supply PP-6419/AWW-4(V) to certain aircraft

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Mounting configurations previously used in mounting the Fuze Function Control Sets AN/AWW-2A and AN/AWW-2B.

3.5.3.2 Weight - The weight of the Adapter Mount MX-8556/AWW-4(V) shall not exceed 1.5 pounds.

3.5.4 Adapter, Mount MX-8557/AWW-4(V) - The Adapter Mount MX-8557/AWW-4(V) shall be manufactured in accordance with the documents listed on DL 67A21D1 and shall meet the requirements stated thereon and in the following subparagraphs.

3.5.4.1 Function - The Adapter Mount MX-8557/AWW-4(V) shall provide the electrical and mechanical interfaces used to adapt the Power Supply PP-6419/AWW-4(V) to certain aircraft mounting and wiring configurations previously used in mounting the Fuze Function Control Set AN/AWW-l.

3.5.4.2 Weight - The weight of the Adapter Mount MX-8557/AWW-4(V) shall not exceed 2.4 pounds.

3.5.5 Adapter, Mount MX-8558/AWW-4(V) - The Adapter Mount MX-85S8/AWW-4(V) shall be manufactured in accordance with the documents listed on DL 67A21D1 and shall meet the requirements stated thereon and in the following subparagraphs.

3.5.5.1 Function - The-Adapter Mount MX-8558/AWW-4(V) shall provide the electrical and mechanical interface used to adapt the Power Supply PP-6419/AWW-4(V) to certain aircraft mounting and wiring configurations previously used in mounting the Fuze Function Control Set AN/AWW-l. The Adapter Mount MX-8558/AWW-4(V) is used when the aircraft is equipped with an integrated cockpit control panel which is wired for use with Fuze Function Control Set AN/AWW-l.

3.5.5.2 Weight - The weight of the Adapter Mount MX-8558/AWW-4(V) shall not exceed 2.4 pounds.

3.5.6 Adapter, Mount MX-9169/AWW-4(V) - The Adapter Mount MX-9169/AWW-4(V) shall be manufactured in accordance with the documents listed on DL 203AS1OO and shall meet the requirements stated thereon and in the following subparagraphs.

3.5.6.1 Function - The Adapter Mount MX-9169/AWW-4(V) shall provide the mechanical interface for mounting the Power Supply PP-6419/AWW-4(V) in the MT-3804/AWW-2A electrical equipment rack. The Adapter Mount MX-9169/AWW-4(V) is used in conjunction with the MT-38O4/AWW-2A electrical equipment rack, in lieu of using the Adapter Mount MX-8556/AWW-4(V), to adapt the Power Supply PP-6419/AWW-4(V) to certain aircraft mounting configurations previously used in mounting Fuze Function Control Sets AN/AWW-2A and AN/AWW-2B when mounting area restrictions do not permit top access to the Power Supply PP-6419/AWW-4(V) mounting screws.

3.5.6.2 Weight - The weight of the Adapter Mount MX-9169/AWW-4(V) shall not exceed 2.8 pounds.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for Inspection - Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the

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Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements.

4.1.1 Classification of Tests - Items covered by this specification shall be subjected to the following tests *to* determine compliance with all applicable requirements :

(1) Preproduction (First Article) Tests

(2) Initial Production Tests

(3) Acceptance Tests

4.2 Preproduction (First Article) Tests - Preproduction tests shall be conducted by the contractor on a minimum of five equipments representative of the production equipments to be supplied under the contract. Preproduction tests shall include all tests delineated in column I of table 1. Preproduction tests shall be accomplished under the approved test procedures of 4.6. The Government Inspector and the procuring activity shall be advised when tests are to be conducted so that a government representative may be designated to witness or supervise the tests when so desired. Contractors not having adequate facilities *to* conduct all required tests shall obtain the services of a commercial testing laboratory acceptable to the Government.

4.2.1 Environmental Test - The operational tests of 4.4.1.2 shall be performed before and after each of the following tests and also during the tests in which MIL-T-5422 specifies equipment operation during testing except that the voltage adjustment of 4.4.1.2.1 shall only be performed prior to the first environmental exposure. Test results shall be forwarded in writing to the procuring activity for review and approval.

4.2.1.1 Temperature/Altitude - This test shall be performed in accordance with MIL-T-5422 (AS) paragraph 4.1 for Class 2 equipment, as required in 3.3.5.l(a).

4.2.1.2 Vibration - This test shall be performed in accordance with MILAT-5422 (AS). G levels of figure 6 will apply. Curves I(a), 5 Hz to 2000 Hz frequency range, will be used in accordance with 3.3.5.l(b).

4.2.1.3 Shock - This test shall be performed in accordance with MIL-T-5422 (AS) paragraph 4.3 as required in 3.3.5.l(a).

4.2.1.4 Humidity - This test shall be conducted in accordance with MIL-T-5422 (AS) paragraph 4.4 as required in 3.3.5. Pre-test and post-test (within 1 hour) data shall be in accordance with 3.3.5.l(d).

4.2.1.5 Salt Fog - This test shall be performed in accordance with MIL-T-5422 (AS) paragraph 4.5 as required in 3.3.5.l(e).

4.2.1.6 Explosion - This test shall be conducted in accordance with MIL-T-5422 (AS) paragraph 4.6, and procedure I will be applicable as required in 3.3.5.l(f).

4,2.1.7 Fungus - This test shall be performed in accordance with MIL-T-5422 (AS) paragraph 4.8 as required in 3.3.5.1(g).

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Table 1. Test Requirements

Requirements Paragraph	Test Paragraph				
	I. Preproduction	II. Acceptance			
		a Individual (see para 4.4.1)	b Sample (see para 4.4.2)	c Reliability Assurance (see para 4.4.3)	d Special (see para 4.4.4)
Performance Characteris-tics 3.4	4.4.101 4.4.1.2 4.2.4	4.4.1.1 4.4.1.2 4.4.1.3	4.4.1.1 4.4.1.2 4.2.4	4.4.1.1 4.4.1.2	4.4.1.1 4.4.1.2 4.2.4
Temp/Altitude 3.3.5.1(a)	4.2.1.1		4.2.1.1		4.2.1.1
Vibration 3.3.5.1(b}	4.2.1.2				
shock 3.3.5.1(C)	4.2.1.3				
Humidity 3.3.5.l(d)	4.2.1.4		4.2.1.4		4.2.1.4
Salt Fog 3.3.5.1(e)	4.2.1.5				
Explosion 3.3.5.1(f)	4.2.1.6				
Fungus 3.3.5.1(g)	4.2.1.7				
Electromagnet c Interference 3.3.3	4.2.2				
Interchange-ability 3.2.2	4.2.4				
Reliability 3,3.2.3				4.4.3	

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4.2.2 Electromagnetic Interference - The Fuze Function Control Set shall be subjected to the tests of MIL-I-6181D and meet the requirements of 3.3.3.

4.2.3 Interchangeability - Interchangeability shall be verified in accordance with 3.2.2.1 of this specification.

4.2.4 Special (Supplemental) Operational Tests - The following tests are required to the extent specified in table 1 and the tests referenced therein.

4.2.4.1 No Load Output Voltage at Extreme Limits of Operating Power Inputs - The Fuze Function Control Set shall meet the requirements of 3.4.3.2 and 3.4.3,3.

4.2.4.2 Recovery Times Under Varying Loads at Extreme Limits of Opcrating Power Inputs - The Fuze Function Control Set shall meet the requirements of 3.4.5.

4.2.4.3 Output Voltage Under Constant Load at Extreme Limits of Operating Power Inputs - The Fuze Function Control Set shall meet the requirements of 3.4.7 with the operating power inputs specified in 3.4.4.1, c and d.

4.2.4.4 Short Circuit at Extreme Limits of Operating Power Inputs - The Fuze Function Control Set shall meet the requirements of 3.4.9.

4.2.4.5 Voltage Drift at Standard Operating Power Input - The Fuze Function Control Set shall meet the VT Position requirements of 3.4.10.1 and INST Position of 3.4.10.2.

4.2.4.6 Resistance of Output - With the pickle circuit de-energized (no HVDC out of 1J2) the d-c resistance between pins 1J2-A and 1J2-B shall be 470 ohms + 15 percent as required in 3.4.14.

4.2.5 Preproduction (First Article) Test Data - The contractor shall submit all data collected in conducting these tests to the procuring activity for review.

4.2.6 Preproduction Approval - Approval of the preproduction sample shall be by the procuring activity upon completion of all tests without any failures. Should any failure occur in the preproduction test, the cognizant NAVAIRSYSCOM project engineer shall review the test results and determine if the preproduction sample should be approved or if additional test samples shall be required. No production equipments shall be delivered prior *to* the approval of the preproduction sample. Prefabrication of production equipment prior to the approval of the preproduction sample is at the contractor's own risk. The approved preproduction sample shall be retained by the contractor for his use in the fabrication and testing of equipment to be submitted for acceptance. The preproduction sample shall not be considered as any of the equipments under the contract.

4.2.7 Production Equipments - Equipments supplied under the contract shall in all respects, including design, construction, workmanship, performance and quality, be equivalent to the approved preproduction sample. Each equipment shall be capable of successfully passing the same tests as imposed on the preproduction sample. Evidence of non-compliance with the above shall constitute cause for rejection of an equipment and of equipment already accepted by the Government. It shall be the obligation of the contractor to make necessary corrections as approved by the procuring activity.

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4.3 Initial Production Tests - When specified in the contract, one of the first 10 production equipments shall be selected and sent at the contractor's expense to a designated Government laboratory for tests. This equipment shall be selected by the procuring activity after the equipment has successfully passed all individual tests. No other tests shall be conducted on the equipment prior to starting the Initial Production Tests. The preproduction sample shall not be selected for this test.

4.3.1 Scope of Tests - This equipment will be subjected to all operational tests (4.4.1.2) and to any environmental tests (4.2.1) the procuring activity deems necessary *to* assure that the production equipment is equivalent to the previously approved preproduction sample in design, construction, workmanship, performance and quality, and that it meets all applicable requirements.

4.3.2 Accessory Material - In addition to the complete equipment submitted for initial production tests the contractor shall also submit such accessory material and data necessary to test the equipment.

4.3.3 Initial Production Sample Approval - Approval of the initial production sample shall be by the procuring activity upon satisfactory completion of all tests. Any design, material, or performance defect made evident during this test shall be corrected by the contractor to the satisfaction of the procuring activity. Failure of the initial production sample to pass any of the tests shall be cause for deliveries of equipment under the contract to cease until proper corrective action is approved and accomplished. Corrective action shall also be accomplished on equipment previously accepted when requested by the procuring activity.

4.3.4 Reconditioning of Initial Production Test Sample - On completion of the initial production test the equipment shall be reworked by the contractor by replacing all worn or damaged items. After reworking, the contractor shall resubmit the equipment for acceptance.

4.4 Acceptance Tests - The contractor shall furnish all samples and shall be responsible for accomplishing the acceptance tests. All inspection and testing shall be under the supervision of the Government Inspector. Contractors not having adequate facilities for conducting all required tests shall engage the service of a commercial testing laboratory acceptable to the procuring activity. The contractor shall furnish test reports showing quantitative results for all acceptance tests. Such reports shall be signed by an authorized representative of the contractor or laboratory, as applicable. Acceptance or approval of material during the course of manufacture shall not be construed as a guarantee of the acceptance of the finished product. Unless otherwise specified by the contract or purchase order, acceptance shall be on a lot basis. A lot shall be defined as the number of Fuze Function Control Sets constructed and submitted for acceptance in 1 month. A Fuze Function Control Set will be accepted only if it successfully passes the individual tests and the lot with which it is associated meets the lot acceptance criteria defined in 4.5.2. Acceptance tests shall consist of the following:

(1) Individual Tests

(2) Sampling Tests

(3) Reliability Assurance Tests

(4) Special Tests

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4.4.1 Individual Tests - Each equipment submitted for acceptance shall be subjected to the individual tests. These tests shall be adequate to determine compliance with the requirements of material, workmanship, operational adequacy, and reliability. As a minimum, each equipment accepted shall have passed the following tests:

(1) Examination of Product

(2) Operational Test

(3) Manufacturing Run-In Test

4.4.1.1 Examination of Product - Each equipment shall be examined carefully to determine that the material and workmanship requirements of MIL-E-5400 and DL 67A24F1 have been met.

4.4.1.2 Operational Test - Each equipment shall be operated long enough to permit the equipment temperature to stabilize and to check sufficient characteristics and record adequate data to assure satisfactory equipment operation. As a minimum the tests of the following paragraphs shall be performed except that the a-c operating power frequency may be 400 Hz nominal.

4.4.1.2.1 No Load Voltage Adjustment at Standard Operating Power Input - The Fuze Function Control Set shall meet the requirements of 3.4.1.

4.4.1.2.2 No Load Voltage at Standard Operating Power Input - The Fuze Function Control Set shall meet all the requirements of 3.4.2.

4.4.1.2.3 No Load Output Voltage at Extreme Limits of A-C Operating Power inputs- The equipment shall meet the requirements of 3.4.3.1 and 3.4.3.4.

4.4.1.2.4 Recovery Time at Standard Operating Power Input Under Varying Loads - The equipment shall meet the requirements of 3.4.4.

4.4.1.2.5 Output Voltage Under Constant Load at Standard Operating Power - Inputs- The equipment shall meet all the requirements of 3.4.6.

4.4.1.2.6 Output Voltage Under Constant Load at Extreme Limits of A-C Operating Power Inputs - The equipment shall meet the requirements of 3.4.7 with the operating power inputs specified in 3.4.7.

4.4.1.2.7 Short Circuit at Standard Operating Power Input - The equipment shall meet the requirements of 3.4.8.

4.4.1.2.8 Interlock Operation - The equipment shall meet the requirements of 3.4.11.

4.4.1.2.9 Pickle Circuit Operation - The equipment shall meet the requirements of 3.4.12.

3.4.1.2.10 High Voltage Indication - The equipment shall meet the requirements of 3.4.13.

4.4.1.3 Manufacturing Run-In Test - Each power supply unit shall be operated under the conditions specified herein for a period of 24 hours without failure. A

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failure shall be defined as anything which causes malfunctioning of the power supply. No adjustments will be permitted during Run-In.

Temperature	Room ambient
Humidity	Room ambient
Vibration	Any selected frequency within the range of 20 to 30 cps (excluding resonant points) and a minimum amplitude of ± 3gts

The Power Supply shall be vibrated (without vibration isolators) for a period of 10 minutes prior to the beginning of the 24 hour operation. The equipment shall be operated during this vibration period for the purpose of detecting flaws and imperfect workmanship. Operation within the specified limits of satisfactory performance is required during the vibration period. The direction of vibration should be vertical to the normal mounting plane for 5 minutes and lateral to the plane for 5 minutes. Where it is not feasible to vibrate the equipment in two directions the vertical direction shall be used. During the 24 hour period of operation following the 10 minute vibration period, the power supply shall be cycled periodically through its various phases of operation. Should a failure occur, it should be repaired and the test started over, except that the 10 minute vibration period need not be repeated when it is certain the failure was not a result of the vibration. Should repetitive failures occur, corrective action shall be taken to eliminate this defect-from future equipment. A record shall be kept of all failures.

4.4.1.3.1 Power Supply Cycling -

(1) The power supply may be constantly left in a -300 volt position or the output voltage selection cycled at the manufacturer's option unless otherwise specified by the procuring activity.

(2) A cycle shall be 10 ± 1 minutes consisting of:

a. A 3 ± .2 minute period during which the pickle circuit is energized. Apply the output voltage of the power supply to a load of 120 ohms + 5 percent in series with 1.0 uf ± 10 percent all in parallel with 10 K ohms ± 5 percent.

b. Immediately after the 3 minute load of (a) the output of the power supply shall be subjected to a short circuit of 2 ± 1 ohms for a duration of 1 to 5 seconds.

c. During the remaining time of the 10 minute cycle the power supply shall be energized and producing high voltage but the pickle circuit shall be de-energized.

4.4.2 Sampling Tests - Equipments selected for sampling tests shall first have passed the individual tests. Equipments shall be selected for sampling tests by the Government Inspector in accordance with the following:

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Size of Lot [See para. 4.4)	Quantity to deselected for Sampling Test
1-50	1
51-125	2
126-225	3
226 and over	4 Fuze Function Control *Sets* plus 1 additional for each additional 200 (or fraction thereof) in the lot.

4.4.2.1 Scope of Tests - As a minimum, each equipment selected for sampling tests shall be subjected *to the* tests listed in column IIb of table 1.

4.4.2.2 Effect of Sample Tests on Lot Disposition - Unless otherwise specified by the contract or purchase order, the lot from which sample test units are selected shall be held pending completion of sample tests. Lot disposition shall be in accordance with 4.5.2.

4.4.3 Reliability Assurance Tests - Reliability Assurance Tests shall be conducted using MIL-STD-781. Tests as required by both the Qualification Phase and the Production Acceptance (Sampling) Phase shall be conducted. Tests shall be conducted on equipments that have passed the Manufacturing Run-In Test 4.4.1.3.

4.4.3.1 Qualification Phase - Prior *to the* acceptance of equipments under the *contract or* order, a minimum of three equipments shall be tested as outlined in MIL-STD-781, under the section entitled "Qualification Phase of Production Reliability Tests". The maximum number of equipments to be used shall be those listed in table 5 of MIL-STD-781. For the qualification phase, Test Level F shall be used. The accept-reject criteria for Test Plan V shall be used.

4.4.3.2 Reliability Sampling Phase Tests - Samples of the equipment shall be tested as outlined in MIL-STD-781 under the section entitled "Sampling Phase of Production Reliability Tests". For the sampling phase, Test Level F shall be used. The accept-reject trite-. for Test Plan V and the requirements *of* 3.3.2.5 shall be used to determine the length of the tests (until an accept or reject decision is reached).

4.4.3.2.1 Lot Size for Sampling Phase - The equipments constructed during 1 month shall be one lot. The Accept-Reject criteria for Test Plan V and the requirements of 3.3.2.5 shall be used to determine the length of the tests in each lot. Unless otherwise specified in the contract or order, no equipments shall be shipped until an Accept decision is reached under each lot.

4.4.3.3 Test Details - The test details such as the length of the test cycle, the length of the heat portion of the cycle, the performance characteristics to be measured, and special failure criteria shall be as specified in this specification.

4.4.3.3.1 The Reliability Sampling Qualification Test shall be conducted on AN/AWW-4(V) units selected from the preproduction sample - not the first production lot as specified in MIL-STD-781.

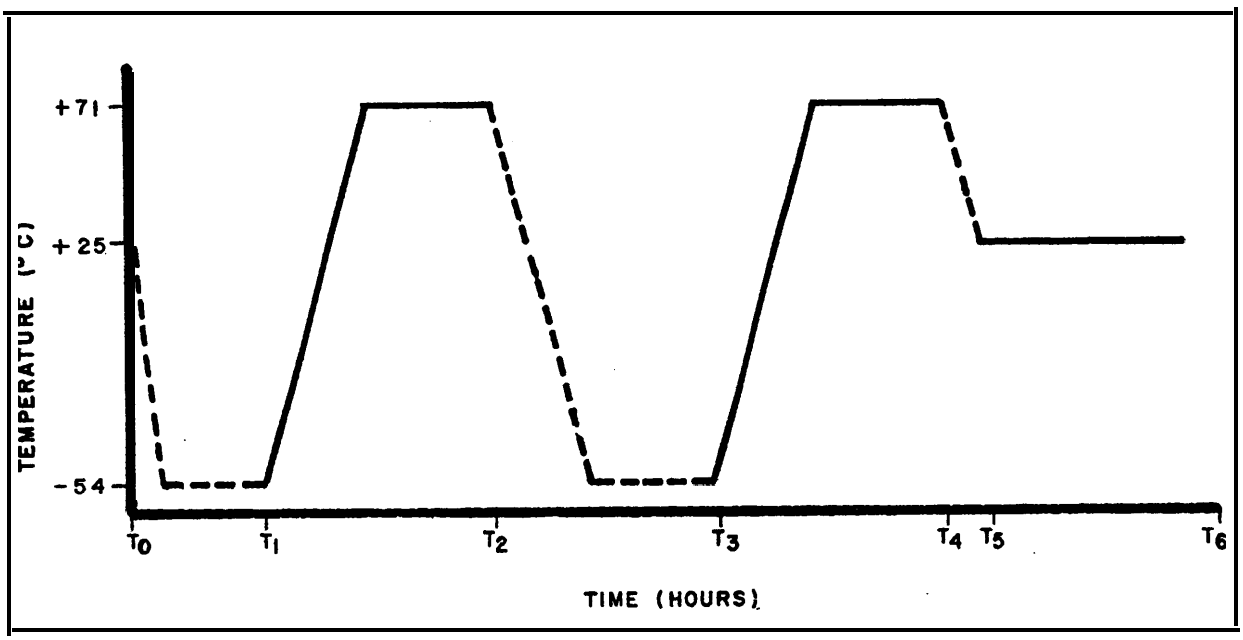
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4.4.3.3.2 The reliability test shall be performed on the preproduction lot and all production lots unless specified by the procuring activity that the test be performed only on the preproduction lot and the first production lot and any other specific lot.

4.4.3.3.3 Vibration frequency shall be between 50 and 60 hertz only.

4.4.3.3.4 The temperature duty cycle shall be as shown by the profile below. The rate of change of the temperature of the test chamber shall be as specified in . MIL-STD-481.

Temperature Time Profile
For
Fuze Function Control Set AN/AWW-4(V)



Legend

-----	Equipment off
____	Equipment on
T_0 T_1	Time for equipment stabilization at -54°C (3 hours)
T_1 - T_2	Equipment on and time for stabilization at +71°C (5 hours)
T_2 - T_3	Equipment off time for stabilization at -54°C (5 hours)
T_3 - T_4	Equipment on and time for stabilization at +71°C (5 hours)
T_4 - T_5	Equipment off and time for stabilization at +25°C (1 hour)

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T_5-T_6 Time to conduct complete Operational Performance Test (5 hours)

T_0-T_6 Twenty-four hour temperature cycle

4.4.3.3.5 Operating Parameters to be Measured at Standard Input - As a minimum, the performance characteristics listed in table 2 shall be measured between times T_5 and T_6 of the Temperature cycle. Failure to meet these requirements shall constitute a failure. During each hour of equipment operation, the units shall be vibrated for 10 minutes at a nonresonant frequency of 50 to 60 Hz with a peak acceleration of 2.0g + 10 percent.

Table 2. Reliability Test Parameters

Requirement	
Output Voltages + and -300 vdc no load limits 3.4.1.2,3.4.202	4.4.1.2.2
Output Voltage +300 vdc 1000 ohm load limits 3.4.6.1	4.4.1.2.5
output Voltages + and -195 vdc no load limits 3.4.2.3, 3.4.2.4.	4.4.1.2.3
Output Voltage +195 vdc 1000 ohm load limits 3.4.6.2	4.4.1.2.5
Short circuit test 3.4.8	4.4.1.2.7
Pickle Circuit Activation Time	3.4.12.1
Pickle Circuit Delay Time	3.4.12.2

4.4.4 Special Tests - Special tests shall be conducted for the purpose of checking the effect of any design or material change on the performance of the equipment and to assure adequate quality control. The equipment selected for special tests may be selected from equipments previously subjected to the sampling or reliability assurance tests.

4.4.4.1 Special Test Schedule - Selection of equipments for special tests shall be made as follows:

(1) On an early equipment after an engineering or material change.

(2) Whenever failure reports or other information indicate additional tests are required. (This will be determinedly the procuring activity.]

4.4.4.2 Scope of Tests - Special tests shall, as a minimum, consist of the tests delineated in Column IId of table 1. Additional tests may be required by the procuring activity based on the nature of failures experienced and/or the changes necessitating the special test. Test procedures previously approved for the

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preproduction tests shall be used where applicable. When not applicable, the contractor shall prepare a test procedure, in accordance with 4.6, and submit it to the procuring activity for approval prior to conducting the special tests.

4.5 Equipment Failure - Should a failure occur during either the sampling, reliability assurance or special tests, the following action shall be taken:

(1) Determine the cause of failure "

(2) Determine if the failure is an isolated case or a design defect.

(3) submit to the procuring activity for approval, proposed corrective . action intended to reduce the possibility of the same failure(s) occurring in future tests.

(4) Where practical, include a test in the individual test to check all equipment for this requirement until reasonable assurance is obtained that the defect has been satisfactorily corrected.

4.5.1 Equipment Failure in Reliability Assurance Testing - When a Fuze Function Control Set failure occurs during reliability assurance tests, the contractor shall follow the procedures specified in MIL-STD-781 applicable to equipment failure.

4.5.2 Lot Disposition - Unless otherwise specified by the procuring activity, a lot will not be accepted unless:

(1) Each Fuze Function Control Set in the lot successfully passes the individual tests of 4.4.1 and

(2) Each Fuze Function Control Set selected from that lot and subjected to sample tests per 4.4.2 successfully passes those tests, and

(3) The lot is accepted per the accept/reject criteria of the reliability assurance tests of 4.4.3 and

(4) The results of any required special tests 4.4.4 indicate compliance with specified performance requirements.

4.5.3 Classification of Characteristics - Inspection of the AN/AWW-4(V) assemblies, prior to and after assembly, shall be in accordance with table 3.

4.6 Test Procedures - The procedures used for conducting preproduction tests, acceptance tests;- and life tests shall be prepared by the contractor and submitted to the procuring activity for review and approval. The right is reserved by the procuring activity or the Government Inspector to modify the tests or require any additional tests deemed necessary. to determine compliance with the requirements of this specification and the contract. MIL-T-18303 shall be used as a guide for preparation of test procedures. When approved test procedures are available from previous contracts such procedures will be provided and may be used when their use is approved by the procuring activity. However, the right is reserved by the procuring activity to require modification of such procedures, including additional tests, when deemed necessary.

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TABLE 30 CLASSIFICATION OF CHARACTERISTICS FOR FUZE FUNCTION CONTROL SET AN/AWW-4(V)

NOTES :

A. This CC covers only those design requirements which affect safety, performance, interchangeability, maintainability, and reliability of items of Government procurement.

B. Conformance with the requirements of this CC does not relieve the contractor of the responsibility for meeting all applicable specification and drawing requirements.

C. In case of conflict between requirements of this CC and subsequent revisions of applicable specifications and drawings, the latest revision of the drawing or specification takes precedence.

D. Sampling and acceptance procedures shall be in accordance with MIL-STD-105, unless otherwise specified in this specification.

E. Reference:

(1) Paragraph numbers are those of this specification.

(2) Drawing numbers are those of DL 125AS1000, DL 126AS100, DL67A21D1, DL 67A22D1, DL 203AS100, DL 67A24F1, and DL 67A35F1.

F. Characteristics, classified as critical or major are recommended for Mandatory Inspection [H-57).

G. Characteristics with numbers followed by the suffix "(S)" (Special), unless covered by a specified sampling plan, shall be verified to the satisfaction of the Government representative.

A. AN/AWIY-4(V] Assembly (Dwg. 67A24F1)

CRITICAL - AQL not defined, 100% inspection, no defects allowed

1. Wheel interlock operation (Ref. para. 3.4.11}

MAJOR - AQL 1.0 percent cumulative for all parameters

101. No load voltage adjustment at standard input (VT) (Ref. para. 3.4.1.1)

102. No load voltage adjustment at standard input (INST) (Ref. para. 3.4.1.2)

103. No load voltage at standard input (VT) (Ref. para. 3.4.2.1)

104. No load voltage at standard input (DLY 2) (Ref. para. 3.4.2.2)

105. No load voltage at standard input (INST) (Ref. para. 3.4.2.3)

106. No load voltage at standard input (DLY 1) (Ref. para. 3.4.2.4)

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TABLE 3. [CONT'D)

A. AN/AWW-4(V) Assembly (Dwg. 67A24F1) (Cont'd)

107.		Recovery time at standard input under varying loads (VT position) (Ref. para. 3.4.4.1)
108.		Recovery time at standard input under varying loads (INST position) (Ref. para. 3.4.4.2)
109.		Recovery time at standard input under varying loads (VT position) (Ref. para. 3.4.4.3)
110.		Output voltage under constant load and standard input (VT position) (Ref. para. 3.4.6.1)
111.		Output voltage under constant load and standard input (INST position) (Ref. para. 3.4.6.2)
112.		Short circuit at standard input (VT position) (Ref. para. 3.4.8.1)
113.		Short circuit at standard input (INST position) (Ref. para. 3.4.8.2)
114.		Voltage drift (VT position) (Ref. para. 3.4.10.1)
115.		Voltage drift (INST position) (Ref. para, 3.4.10.2]
116.		High voltage indication (+300 volt position) (Ref. para. 3.4.13)
117.		High voltage indication (+195 volt position) (Ref. para. 3.4.13)
118.	(S)	No-load output voltage with varying voltage and frequency input (VT position) (Ref. para. 3.4.3.1.a)
119.	(s)	DLY 2 position (Ref. para. 3.4.3.1.b)
120.	(s)	INST position (Ref. para. 3.4.3.1.c)
121.	(s)	DLY 1 position (Ref. para. 3.4.3.l.d)
122.	(s)	VT position (Ref. para. 3.4.3.2 and 3.4.3.1.a)
123.	(S)	DLY 2 position (Ref. para. 3.4.3.2 and 3.4.3.1.b)
124.	(S)	INST position (Ref. para. 3.4.3.2 and 3.4.3.1.c)
125.	(S)	DLY 1 position (Ref. para. 3.4.3.2 and 3.4.3.1.d)
126.	(S)	VT position (Ref. para. 3.4.3.3 and 3.4.3.1.a)
127.	(S]	DLY 2 position (Ref. para. 3.4.3.3 and 3.4.3.1.b)
128.	(S)	INST position (Ref. para. 3.4.3.3 and 3.4.3.1.c)

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TABLE 3. (CONT'D)

A. AN/AWW-4(V) Assembly (Dwg. 67A24F1) (Cont'd)

129.	(S)	DLY 1 position (Ref. para. 3.4.3.3 and 3.4.3 .1. d)
130.	(s)	VT position (Ref. para. 3.4.3.4 and 3.4.3.1.a)
131.	(s)	DLY 2 position (Ref. para. 3.4.3.4 and 3.4.3.1.b)
132.	(S)	INST position (Ref. para. 3.4.3.4 and 3.4.3.1.c)
133.	(s)	DLY 1 position (Ref. para. 3.4.3.4 and 3.4.3.1.d)
134.	(s)	Recovery times under varying loads at extreme limits of operating power input (VT position) (Ref para. 3.4.4.1.a and 3.4.3.1)
135.	(s)	INST position (Ref. para. 3.4.4.1.a and 3.4.3.2)
136.	(S)	VT position (Ref. para. 3.4.4.1.a and 3.4.3.3)
137.	(s)	VT position (Ref. para. 3.4.4.1.b and 3.4.3.1)
138.	(S)	INST position (Ref. para. 3.4.4.1.b and 3.4.3.2)
139.	(s)	VT position (Ref. para. 3.4.4.1.b and 3.4.3.3)
140.	(s)	VT position (Ref. para. 3.4.4.1.c and 3.4.3.1)
141.	(s)	INST position (Ref. para 3.4.4.1.c and 3.4.3.2)
142.	(S)	VT position (Ref. para. 3.4.4.1.c and 3.4.3.3)
143.	(s)	VT position (Ref. para. 3.4.4.1.d and 3.4.3.1)
144.	(s)	INST position (Ref. para. 3.4.d.1.d and 3.4.3.2)
145.	(s)	VT position (Ref. para. 3.4.A.1.d and 3.4.3.3)
146.	(S)	Output voltage under constant load and with varying. voltage and frequency (VT position) (Ref. para. 3.4.6.1, 3.4.5.1, and 3.4.4.1.a)
147.	(s)	INST position (Ref. para. 3.4.6.1, 3.4.5.2, and 3.4.4.1.a)
148.	(S)	VT position (Ref. para. 3.4.6.1, 3.4.5.1, and 3.4.4.1.b)
149.	(s)	INST position (Ref. para. 3.4.6.1, 3.4.5.2, and 3.4.4.1.b)
150.	(s)	VT position (Ref. para. 3.4.6.1, 3.4.5.1, and 3.4.4.1.c)
151.	[s)	INST position (Ref. para. 3.4.6.1, 3.4.5.2, and 3.4.4.1.c)
152.	(s)	VT position (Ref. para. 3.4.6.1, 3,4.5.1, and 3.4.4.1.d)

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TABLE 3. (CONT'D)

A. AN/AWW-4(V) Assembly (Dwg. 67A24F1) (Cont'd)

153. (S] INST position (Ref. para. 3.4.6.1, 3.4.5.2, and 3.4.4.1.d]

154. (S) Short circuit at varying voltage and frequency (VT position) (Ref. para 3.4.8.1, 3.4.7.1, and 3.4.4.1.a)

155. (S) INST position (Ref. para. 3.4.8.1, 3.4.7.2, and 3.4.4.1.a)

156. (S) VT position (Ref. para. 3.4.8.1, 3.4.7.1, and 3.4.4.1.b)

157. (S) INST position (Ref. para. 3.4.8.1, 3.4.7.2, and 3.4.4.1.b)

158. (S) VT position (Ref. para. 3.4.8.1, 3.4.7.1, and 3.4.4.1.c)

159. (S) INST position (Ref. para. 3.4.8.1, 3.4.7.2, and 3.4.4.1.c}

160. (S) VT position (Ref. para. 3.4.8.1, 3.4.701, and 3.4.4.l.d)

161. (S) INST position (Ref. para. 3.4.8.1, 3.4.7.2, and 3.4.4.l.d)

162. Cable connectors on power supply must be mounted properly and pins not bent, broken or loose (Ref. Dwg. 125AS1000]

163. Cable connectors on control unit must be mounted correctly and not bent, broken or loose (Ref. Dwg. 126AS100)

164. Clearance must be as specified (Ref. Dwg. 67A24F1)

165. All connectors must mate (Ref. Dwg. 67A24F1) .

166. Circuit Breakers and lamp must be present and workable (Ref. Dwg. 125 AS1000 and 126AS100)

167. All lettering must be clear, legible and correct (Ref. Dwg. 125AS100 and 126AS100)

168. Elapsed time indicator must be present (Ref. Dwg. 125AS1000)

169. All threaded surfaces must be free from nicks, burrs and dust (Ref. Dwg. 125AS1000 and 126AS100)

170. Dimensions and weights of Power Supply/Adapter Assemblies must be as specified (Ref. Dwg. 67A24E10, 67A24E11, 67A24E12, and 67A24E13)

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TABLE 3. (CONT'D)

B. FINAL ASSEMBLY, POWER SUPPLY PP6419/AWW-4(V) (DWG. 125AS1000)

CRITICAL - None

MAJOR - AQL 1.0 cumulative for all parameters

101. Use thread sealing compound as specified (Ref. Dwg. 125AS1000)

102. Route wire bundles as specified (Ref. Dwg. 125AS1000)

103. Install washers and sleeving as specified (Ref. Dwg. 125AS1000)

104. Wiring and soldering materials and procedures must be as specified (Ref. Dwg. 125AS1000]

105. Wire lacing must be as specified (Ref. Dwg. 125AS1000)

106. Application of compound to heat sink must be as specified (Ref. Dwg. 125AS1000)

107. Dimensions and weight must be as specified (Ref. Dwg. 125AS104)

108. Wiring must be as specified (Ref. Dwg. 125AS102)

109. Power transformer must be as specified (Ref. Dwg. 125AS112)

110. Power supply inductor must be as specified (Ref. Dwg. 125AS122) ,

111. Wiring, soldering and sleeving of relay assembly must be as specified {Ref. Dwg. 125AS123)

112. Identification plate dimensions and requirements must be as specified (Ref. Dwg. 12SAS124)

113. Ground cable assembly must be as specified (Ref. Dwg. 125AS127)

C. FINAL ASSEMBLY, CONTROL C-8441 and C-8442/AWW-4(V) (DWG. 126AS100)

CRITICAL - None

MAJOR - AQL 1.0 percent cumulative for all parameters

101. Wiring and soldering of electrical connections must be as specified (Ref. Dwg. 126AS100)

102. Lacing, sleeving and varnish must be as specified (Ref. Dwg. 126AS100)

103. Black oxide coating application must be as specified (Ref. Dwg. 126AS100)

104. Panel lighting for all applications must be as specified (Ref. Dwg. 126AS100)

105. Lock washers to be as specified (Ref. Dwg. 126AS100)

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TABLE 3. (CONT'D)

c. FINAL ASSEMBLY , CONTROL C-8441 and C-8442 /AWW-4 (V) (DWG. 126AS100) (Cont'd)

106. Item 11 is to clear B by 0.010 inch (Ref. Dwg. 126AS100)

107. Dimensions and weight must be as specified (Ref. Dwg. 126AS104)

108. Aircraft power requirements and control must be as specified (Ref. Dwg. 126AS104)

109. Voltage marking on edge of panel must be as specified (Ref. Dwg. 126AS104)

110. Illuminated knob pointer dimensions and testing shall be as specified (Ref. Dwg. 126AS111)

111. Identification plate dimensions and requirements shall be as specified (Ref. Dwg. 126AS111)

112. Control cover assembly dimensions and requirements shall be as specified (Ref. Dwg. 126AS114)

113. Mounting plate assembly dimensions and requirements shall be as specified (Ref. Dwg. 126AS115)

114. Connector plate dimensions and requirements shall be as specified (Ref. Dwg. 126AS118)

115. Illuminating panel dimensions and requirements must be as specified (Ref. Dwg. 126AS120)

D. FINAL ASSEMBLY, ADAPTER, MOUNT MX-8557 and MX-8558/AWW-4(V) (DWG. 67A21D1-1 and 67A21D1-2)

CRITICAL - None

MAJOR - AQL 1.0 percent cumulative for all parameters

101. Wiring and soldering of electrical connections must be as specified (Ref. Dwgs, 67A21D6, 67A21D7, 67A21C8, 67A21D2)

102. Adapter shall mate with Power Supply PP-6419/AWW-4(V) (Ref. Dwgs. 67A21D4, 67A27E10, 67A27E11, 125AS104)

103. Mounting plate dimensions shall be-as specified (Ref. Dwg. 67A21D4)

104. Connector Bracket dimensions shall be as specified (Ref. Dwgs. 67A21C5, 67A21C15)

105. Identification plate dimensions and requirements shall be as specified (Ref. Dwg. 67A21C9)

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TABLE 3. (CONT'D)

E. FINAL ASSEMBLY, ADAPTER, MOUNT MX-8556/AWW-4(V) (DWG. 67A22D1)

CRITICAL - None

MAJOR - AQL 1.0 percent cumulative for all parameters

101. Dimensions shall be as specified (Ref. Dwg. 67A22D2)

102. Adapter shall mate with Power Supply PP-6419/AWW-4(V) (Ref. Dwgs. 67A22D2, 125AS104, 67A24E12)

103. Identification plate dimensions and requirements shall be as specified (Ref. Dwg. 67A22C4)

F. FINAL ASSEMBLY, ADAPTER, MOUNT MX-9169/AWW-4(V) (DWG. 203AS100)

CRITICAL - None

MAJOR - AQL 1.0 percent cumulative for all parameters

101. Dimensions of mounting plate and spacers shall be as specified (Ref. Dwgs. 203AS101 and 203AS103)

102. Dimensions of mounting brackets shall be as specified (Ref. Dwg. 203AS102)

103. Identification plate dimensions and requirements shall be as specified (Ref. Dwg. 203AS104)

104. Spacers must be secured to mounting plate (Ref. Dwg. 203AS101)

105. Adapter shall mate with Power Supply PP-6419/AWW-4(V) and Equipment Rack MT-3804/AWW-2A (Ref. Dwgs. 67A35F1, 203AS104, 125AS104, 67A24E13)

4.7 Reconditioning of Test Equipment - Equipment which has been subjected to acceptance and life tests shall be reconditioned by the contractor by replacing all wear or damaged items. After reworking, the contractor shall resubmit the equipment for acceptance.

4.8 Presubmission Testing - No item, part or complete equipment shall be submitted by the contractor until it has been previously tested and inspected by the contractor and found *to* comply, to the best of his knowledge and belief, with all applicable requirements.

4.9 Rejection and Retest - Equipment which has been rejected may be reworked or have parts replaced *to* correct the defects and resubmitted for acceptance. Before resubmitting, full particulars concerning previous rejection and the action taken to correct the defects found in the original shall be furnished to the Government Inspector.

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5. PREPARATION FOR DELIVERY

5.1 General - All major units and parts of the equipment shall be preserved, packaged, packed and marked for the level of shipment specified in the contract or order in accordance with MIL-E-17555, In the event the-equipment is not covered in MIL-E-17555, the method of preservation for Level A shall be determined in accordance with the selection chart in Appendix D of MIL-STD-794.

60 NOTES

6.1 Intended Use - The Fuze Function Control Set AN/AWW-4(V) is used to provide a means of selecting arming delay times, firing delay times, and energizing electric weapon fuzes.

6.2 Performance Objectives - Minimum size and weight, simplicity of operation, ease of maintenance, and an improvement in the performance 'and reliability of the specific functions beyond the requirements of *the* specification are objectives which shall be considered in the production of this -equipment. Where it appears a substantial reduction in size and weight or improvement in simplicity of design, performance, ease of maintenance or reliability will result from the use of materials, parts and processes other than those specified in DL 67A24F1 and related lists, it is desired that their use be investigated. When investigation shows advantages can be realized, a request for approval shall be submitted to the procuring activity for consideration. Each request shall be accompanied by complete supporting information.

6.3 Precedence of Documents - When the requirements of the contract, this specification, or applicable subsidiary specifications are in conflict, the fallowing precedence shall apply:

(1) Contract - The contract shall have precedence over any specification.

(2) This Specification - This specification shall have precedence over all applicable subsidiary specifications. Any deviation from this specification, or horn subsidiary specifications where applicable, shall be specifically approved in writing by the procuring activity.

(3) The Drawings of DL 67A24F1 and Related Lists - The drawings shall have precedence over other referenced specifications. Any deviation from the drawings shall be specifically approved by the procuring activity.

(4) Referenced Specifications - Any referenced specifications shall have precedence over all applicable subsidiary specifications referenced therein. All referenced specifications shall apply *to* the extent specified.

6.4 Ordering Data - Purchasers should exercise any desired options offered herein, and procurement documents should specify the following:

(1) Title, number, and date of this specification.

(2) Selection of applicable levels of packaging and packing (see 5.1).

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(3) If Initial Production Tests of 4.3 are required and if so the designated Government laboratory.

(4) If Reliability Sampling Phase Tests of 4.4.3.2 are to be deleted and, if so which lots.

6.5 Asterisk - In specification revisions and superseding amendments, an asterisk "*" preceding a paragraph number denotes paragraphs in which changes have been made from the previous issue. This has been done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations, Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content as written irrespective of the asterisk notations and relationship to the last previous issue.

6.6 Associated Equipment - The equipment shall operate with the following associated equipment:

(1) Fuze Function Control Test Set AN/AWM-42A

(2) Test Interconnecting Kit MK-1250/AWM-42

(3) Electrical Equipment Rack MT-3804/AWW-2A

Project No. 1290-NS22

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

OMB Approval
No. 22-R255

INSTRUCTIONS: The purpose of this form is to solicit beneficial comments which will help achieve procurement of suitable products at reasonable cost and minimum delay, or will otherwise enhance use of the document. DoD contractors, government activities, or manufacturers/vendors who are prospective suppliers of the product are invited to submit comments to the government. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements. Attach any pertinent data which may be of use in improving this document. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity.

DOCUMENT IDENTIFIER AND TITLE
MIL-F-81966(AS) FUZE FUNCTION CONTROL SET AN/AWW-4(V)

NAME OF ORGANIZATION AND ADDRESS	CONTRACT NUMBER
	MATERIAL PROCURED UNDER A ☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?
   A. GIVE PARAGRAPH NUMBER AND WORDING.

   B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY DOCUMENT REQUIREMENT CONSIDERED TOO RIGID

3. IS THE DOCUMENT RESTRICTIVE?
   ☐ YES ☐ NO *(If "Yes", in what way?)*

4. REMARKS

SUBMITTED BY *(Printed or typed name and address - Optional)*	TELEPHONE NO.
	DATE

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